Experiment 3

Ex. No.: 3

Recover Deleted Files Using Forensics Tools

Aim

To understand how a deleted test file can be recovered from a forensic image without altering the original storage medium.

Tools required

1. A disposable flash drive or synthetic test image containing a harmless text file.
2. Notepad or another text editor for creating the sample file.
3. FTK Imager for bit-for-bit acquisition (the manual workflow).
4. RecoverMyFiles for recovery from the image (the manual workflow).
5. The dd command may be used as an imaging alternative on Linux when the source is correctly identified and protected.

Idea/Theory

1. Deleting a file usually removes its directory reference before its data blocks are overwritten.
2. Ordinary file copies omit unallocated space and deleted entries; a forensic image preserves the addressable source sectors for examination.
3. Recovery tools scan file-system metadata and file signatures to locate recoverable content.
4. The examiner should hash the image, work from a copy, and avoid writing to the source device.

Experiment 3 — continued

Procedure/Steps

1. On a disposable lab medium, create a harmless text file and record its name and contents.
2. Delete the file through the operating system and, where required by the lab, empty the recycle bin.
3. Attach the medium through a write blocker when available. In FTK Imager choose **File → Create Disk Image**.
4. Enter the case number, evidence number, examiner name, source, destination, and image name requested by the wizard.
5. Create and verify the bit-for-bit image. A Linux alternative is shown below; the source must be read-only and positively identified before it is used.

```
dd if=<read-only-source> of=evidence.img bs=4M conv=noerror,sync
```

> Do not substitute a personal drive for the disposable lab medium.

Experiment 3 — continued

Procedure/Steps (continued)

1. Open RecoverMyFiles and select **Start Recovery**.
2. Choose **Recover a Drive**, then use **Add Image** to select the FTK image rather than the original medium.
3. Start automatic recovery and wait for the scan to complete.
4. Use the **File Type** view to filter for the deleted file type, such as TXT.
5. Preview the candidate, export it to a separate destination, and compare its content and hash with the known sample.

Observation/Output

Screenshots to be attached from the lab run (tool: FTK Imager and RecoverMyFiles)

Execution note: Sleuth Kit, Foremost, PhotoRec, FTK Imager, and RecoverMyFiles were not installed in this environment, so no recovery output was invented.

Result

To be completed after the lab run.